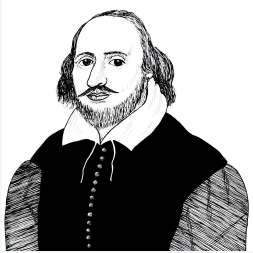
AI as Engineering Tool Experiment

SWEaters United - Team 02
Model: Gemini 3.1 Pro Preview

GEMINI

Research



- Before development we researched on origin, history, regulations, key features, competing apps, and more on slot machines

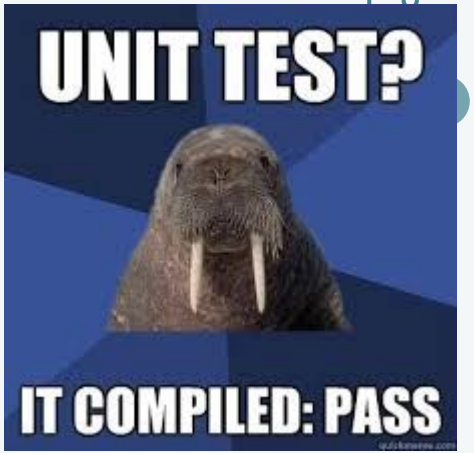
- We settled on key features and bonus features we wanted in our slot machine using our research

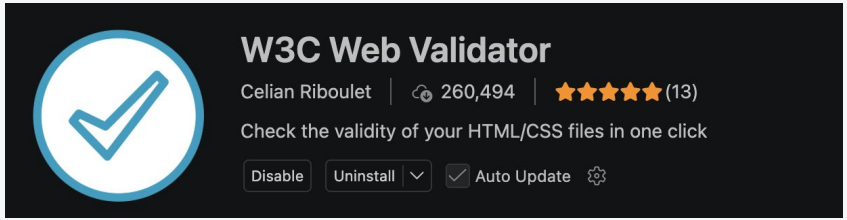
- Also came up with two personas + seven user stories which helped give more context to the agent

Instructions
Index
Research
Definition
Terminology
Regulations
User Personas
User Stories
Slot Machine Apps & Featu...
Visual Themes
Case Studies/Website
History and Origin
Feature Gap Analysis
Odds/Payout Specifics

Code standards + Linting/Testing

- Gave agent clear code standards before any other prompts
- Asked agent to check its code via the linter/unit tests it created after each improvement
- *But*.... we can't trust agents: we also linted and ran tests ourselves, and also used a HTML/CSS W3C validator extension
- We found that the agent followed these standards, but needed to be reminded as we did more iterations





Development Phases

- **Phase 0**: Inform AI of goal/background knowledge
  - No files created. Only show our desired slot machine.
  - Specify it to use HTML, CSS, and JS
- **Phase 1**: Understanding the user and personas
  - Tell the AI to acknowledge all user personas and stories
- **Phase 2**: Core Game Engine (JS)
  - Work on game logic only
  - Symbol definitions, Coin balance math, Win condition, …
- **Phase 3**: Basic UI (HTML)
  - Build basic interface structure
  - Connect with JS file
- **Phase 4**: Styling/Game feel (CSS)
  - Make sure AI only work on CSS
  - Define theme and vibe
- **Phase 5**: Bonus features
  - Wilds, scatter, free spins, …

Phase 4
(run 11 -> 12)

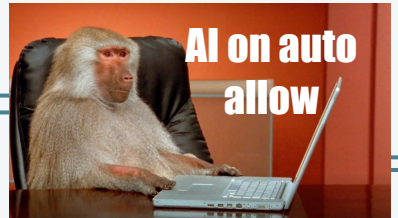
Observations from Runs

- Multiple times, the AI model did not function as intended
- AI works best with divided, simpler tasks one by one
- It's hard to determine what's missing without a styled website
- A lot of compliance and variance drift
- Sometimes there are lucky runs
- Performs worse on auto allow / when thinking for extended time periods

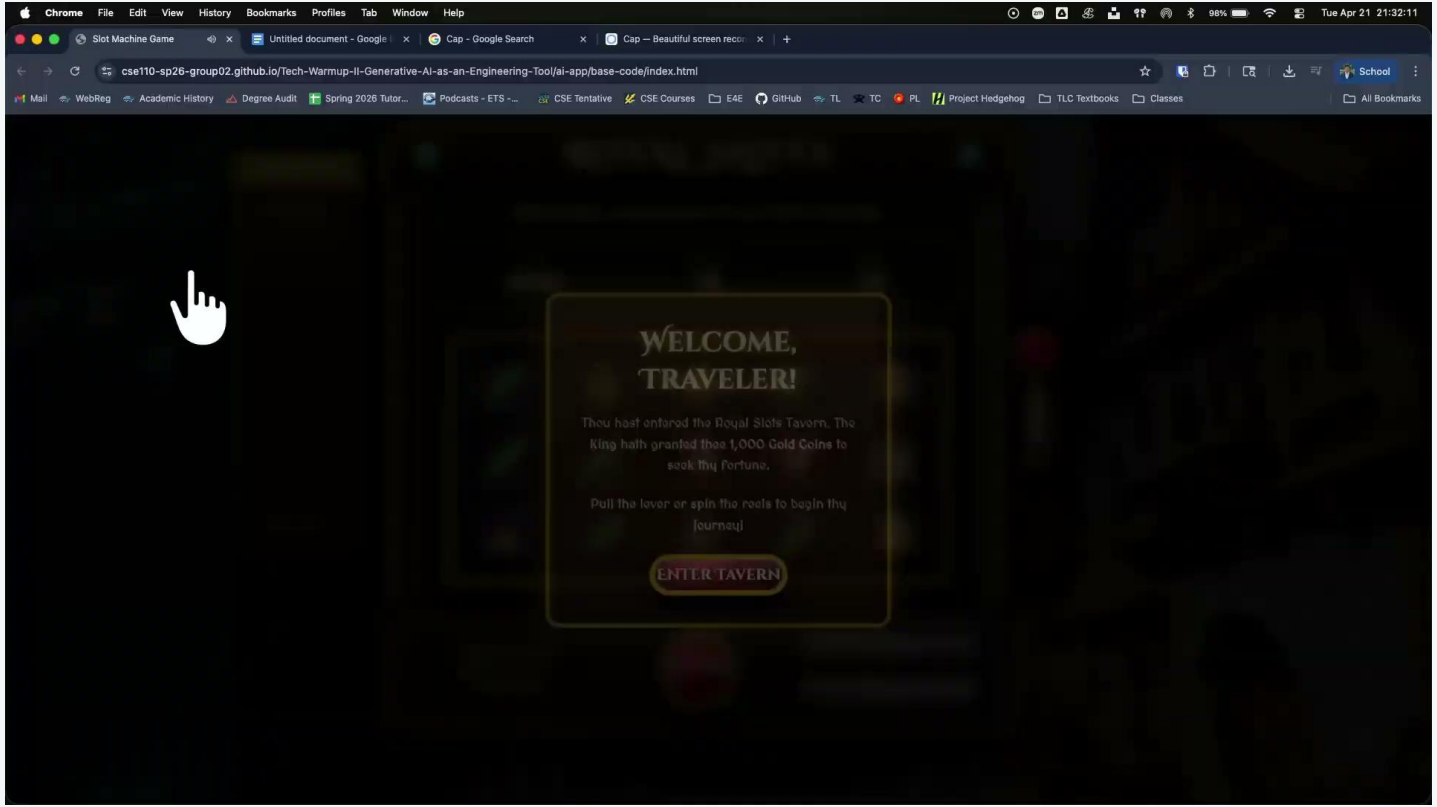
Sizing issues (run 17)

```
> npm run lint:js


> tech-warmup-ii-generative-ai-as-an-engineering-tool@1.0.0 lint:js
> npx eslint "ai-app/base-code/**/*.js"


C:\Users\Scott\Desktop\school\cse110\Tech-Warmup-II-Generative-AI-as-an-Engineering-Tool\ai-app\base-code\ui-controller.js
    9:47  warning  Arrow function has too many lines (343). Maximum allowed is 50   max-lines-per-function
   40:49  warning  No magic number: 100                                              no-magic-numbers
   43:28  warning  No magic number: 1000                                             no-magic-numbers
   44:25  warning  No magic number: 10                                               no-magic-numbers
  109:5   warning  Function 'renderGrid' has too many lines (53). Maximum allowed is 50   max-lines-per-function
  109:5   warning  Function 'renderGrid' has a complexity of 9. Maximum allowed is 5     complexity
  189:5   warning  Async function 'executeSpin' has too many lines (51). Maximum allowed is 50   max-lines-per-function
  189:5   warning  Async function 'executeSpin' has a complexity of 6. Maximum allowed is 5     complexity
  189:40  warning  No magic number: 5                                               no-magic-numbers
  203:54  warning  No magic number: 0.15                                            no-magic-numbers
```

Big prompts with many features = worse code quality (run 8)



Video Demo


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Tue Apr 21 21:32:11
Slot Machine Game
Untitled document - Google
Cap - Google Search
cse110-sp26-group02.github.io/Tech-Warmup-II-Generative-AI-as-an-Engineering-Tool/ai-app/base-code/index.html
WELCOME, TRAVELER!
Thou hast entered the Royal Slots Tavern. The King hath granted thee 1,000 Gold Coins to seek thy fortune.
Pull the lever or spin the reels to begin thy journey!
ENTER TAVERN

Final Thoughts

What Worked Well

- Code runs without error
- Good Priming
- Significant Complexity Reduction
- Self-Correction

Significantly Better Features

- Has great audio
- Able to cash out with a receipt to verify
- Correct Calculations

What Didn't Work Well

- Complex Prompt
- Lack of Visual Feedback
- Simultaneous Styling and Structure
- Finding Valid Links to External Resources
- Specific UI Positioning

Was Code Easy to Understand?

- Good format
- A lot of
- No comments
- Organized structure
- Still has room for improvement

Not a reliable way to produce a software!